

MS Algebra Recitation 4

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1 The First Isomorphism Theorem Revisited

Let V and W be vector spaces over a field k . Then V and W are in particular abelian groups. If $T : V \rightarrow W$ is linear, then it is in particular a homomorphism of abelian groups. Thus, $\ker T \leq V$, and so the first isomorphism theorem gives us that $V/\ker T \cong \text{im } T$. It actually turns out that $\ker T$ is a *subspace* of V , and that this isomorphism holds as vector spaces. Taking dimensions,

$$\dim V - \dim \ker T = \dim(V/\ker T) = \dim \text{im } T.$$

So, the first isomorphism theorem gives us back rank-nullity.

2 The Universal Property of Quotients

Let's show how to construct a map out of a quotient group.

Theorem (Universal Property of Quotients). *Let G be a group and $N \trianglelefteq G$ be normal. Suppose that $\phi : G \rightarrow H$ is a homomorphism with $\phi(N) = 1$. Then there exists a unique map $\Phi : G/N \rightarrow H$ such that the following diagram commutes*

$$\begin{array}{ccc} G & \xrightarrow{\phi} & H \\ \pi \downarrow & \nearrow \exists! \Phi & \\ G/N & & \end{array}$$

That is, there is a unique Φ such that $\Phi \circ \pi = \phi$.

Proof. For such a Φ to exist, we must have $\Phi(gN) = \Phi(\pi(g)) = \phi(g)$. So, we must have uniqueness. This is immediately a homomorphism:

$$\Phi(gN \cdot hN) = \Phi(ghN) = \phi(gh) = \phi(g)\phi(h) = \Phi(gN)\Phi(hN).$$

Thus, we only need to check that it is well-defined. If $gN = hN$, then $h^{-1}g \in N$, so that $\phi(h^{-1}g) = 1$. Thus, $\phi(g) = \phi(h)$, so it doesn't matter what representative we pick and Φ is well-defined. ■

Citing this theorem is going to be very useful for constructing maps in general.

3 The Abelianization

The point of this section is to construct the “most general” abelian group from any group. So let's first think about what we would mean by “most general”.

Example. Let X be a metric space. We have a *complete* metric space $\overline{X}$ with an inclusion $X \hookrightarrow \overline{X}$ (this is the set of Cauchy sequences in X under the metric $d((x_n)_n, (y_n)_n) = \lim_{n \rightarrow \infty} d(x_n, y_n)$ - prove this is complete if you've not seen it before). This is considered to be the “smallest” complete metric space containing X for the following reason: if Y is a complete metric space and we have an inclusion $X \hookrightarrow Y$, this induces an inclusion $\overline{X} \hookrightarrow Y$. In fact, this is still true if we have any uniformly continuous $X \rightarrow Y$.

Example. Let $X \subseteq \mathbb{R}^n$ and let $E \subseteq \mathbb{R}^n$ be closed. If $X \subseteq E$, then we have that $\overline{X} \subseteq E$ (where $\overline{X}$ is the closure of X).

In both of these examples, we have a map from our X to some object that has our desired property P , and claim that the “most general version of X having P ” has an induced map into that object. So we'll use this as our definition (note that the universal property of quotients gives us a similar result).

Definition. Let G be a group and $g, h \in G$. We define their *commutator* as

$$[g, h] = ghg^{-1}h^{-1}.$$

The *commutator subgroup* of G is

$$[G, G] = \langle [g, h] : g, h \in G \rangle.$$

Note that g and h commute if and only if $[g, h] = 1$, and so in particular $[g, h] = 1$ for all $g, h \in G$ whenever G is abelian. Note also that

$$\phi([g, h]) = \phi(ghg^{-1}h^{-1}) = \phi(g)\phi(h)\phi(g)^{-1}\phi(h)^{-1} = [\phi(g), \phi(h)]$$

whenever ϕ is a homomorphism out of G . In particular, if $\phi : G \rightarrow H$ with G abelian, we have that $\phi([G, G]) = 1$, since ϕ sends all the generators to 1. This is almost the requirements universal property of quotients, we just need the following:

Proposition. Let G be a group. Then $[G, G] \text{ char } G$. In particular, $[G, G] \trianglelefteq G$.

Proof. Let $\sigma : G \rightarrow G$ be an automorphism. Then $\sigma([g, h]) = [\sigma(g), \sigma(h)] \in [G, G]$. Thus, $\sigma([G, G]) \leq [G, G]$, since the image of every generator is contained in $[G, G]$. This same argument applied to σ^{-1} shows that $\sigma^{-1}([G, G]) \leq [G, G]$. Since σ is a bijection, this gives us $[G, G] \leq \sigma([G, G])$. ■

In particular, the following is well-defined.

Definition. Let G be a group. Its *abelianization* is $G^{ab} = G/[G, G]$.

Note that the abelianization is abelian, since (writing $N = [G, G]$) $[gN, hN] = [g, h]N = N$. So putting together all our observations gives the following:

Theorem (Universal Property of Abelianization). Let G be a group and H be an abelian group. For any homomorphism $\phi : G \rightarrow H$, there exists a unique $\Phi : G^{ab} \rightarrow H$ such that the following diagram commutes

$$\begin{array}{ccc} G & \xrightarrow{\phi} & H \\ \pi \downarrow & \nearrow \exists! \Phi & \\ G^{ab} & & \end{array}$$

It actually turns out that G^{ab} is unique, in the following sense.

Proposition. Suppose we have an abelian group $G^{ab'}$ with a map $\pi' : G \rightarrow G^{ab'}$ satisfying the same universal property. then $G^{ab} \cong G^{ab'}$.

Proof. Since $G^{ab'}$ is abelian, the universal property of G^{ab} gives us a map $\Pi' : G^{ab} \rightarrow G^{ab'}$. Similarly, the universal property of $G^{ab'}$ gives us a map $\Pi : G^{ab'} \rightarrow G^{ab}$. Thus, we have the following commutative diagram:

$$\begin{array}{ccc} G & \xrightarrow{\pi'} & G^{ab'} \\ \pi \downarrow & \nearrow \Pi' & \\ G^{ab} & \xleftarrow{\Pi} & \end{array}$$

Now,

$$\Pi \circ \Pi' \circ \pi = \Pi \circ \pi' = \pi,$$

so that $\Pi \circ \Pi' = \text{id}$ since π is surjective. Similarly $\Pi' \circ \Pi = \text{id}$ since π' is surjective. So we've found an isomorphism between G^{ab} and $G^{ab'}$. ■

Currently, we have a way to abelianize groups. It turns out we can also abelianize maps (this makes abelianization into a *functor*).

Proposition. Let G and H be groups and $\phi : G \rightarrow H$ be a homomorphism. Then there is a unique $\phi^{ab} : G^{ab} \rightarrow H^{ab}$ such that the following diagram commutes

$$\begin{array}{ccc} G & \xrightarrow{\phi} & H \\ \pi_G \downarrow & & \downarrow \pi_H \\ G^{ab} & \xrightarrow{\exists! \phi^{ab}} & H^{ab} \end{array}$$

Proof. We have a map $G \rightarrow H \rightarrow H^{ab}$ from G into an abelian group, and so the universal property of abelianization gives us a unique $\phi^{ab} : G^{ab} \rightarrow H^{ab}$. ■

4 The Symmetric Group

We'll investigate the automorphisms of the symmetric group S_n . Let's first look at the structure of the group.

Proposition. S_n is generated by

- (a) The 2-cycles
- (b) The set $\{(12), (13), \dots, (1n)\}$

Proof.

- (a) Since any permutation is the product of disjoint k -cycles, it suffices to show that a k -cycle is a product of 2-cycles. We have that

$$(i_1, i_2, \dots, i_k) = (i_1, i_2)(i_2, i_3) \cdots (i_{k-1}, i_k).$$

- (b) It suffices to show that 2-cycles are a product of these elements. We have that

$$(1j)(1i)(1j) = (ij).$$

■

The following gives us a first grasp on automorphisms of S_n .

Proposition. Let $n \geq 3$. Then S_n is isomorphic to a subgroup of $\text{Aut}(S_n)$.

Proof. For any group G , the map $G \rightarrow \text{Aut}(G)$ by $g \mapsto \varphi_g$ is a homomorphism with kernel $Z(G)$. So it suffices to show that $Z(S_n) = 1$.

Let $\sigma \in S_n$ be a nonidentity permutation, say $\sigma(i) \neq i$. Since $n \geq 3$, we can take $j \neq i, \sigma(i)$. But then

$$((ij) \circ \sigma \circ (ij))(j) = \sigma(i)$$

while $\sigma(j) \neq \sigma(i)$. Thus, σ doesn't commute with (ij) . ■

So what we will investigate is the index $[\text{Aut}(S_n) : S_n]$.

Lemma. Let G be a group and σ be an automorphism of G . Then σ acts on the set of conjugacy classes of G . That is, if $\mathcal{C} \subseteq G$ is a conjugacy class of G , then $\sigma(\mathcal{C})$ is also a (full) conjugacy class of G .

Proof. Fix a conjugacy class $\mathcal{C}$, and take $a, b \in \mathcal{C}$ with $a = gbg^{-1}$. Then $\sigma(a) = \sigma(g)\sigma(b)\sigma(g)^{-1}$, so $\sigma(a)$ and $\sigma(b)$ lie in the same conjugacy class of G .

We claim that $\sigma(\mathcal{C})$ must be this whole conjugacy class. Suppose we have some other d conjugate to $\sigma(a)$, say $d = h\sigma(a)h^{-1}$. Since σ is an automorphism, we can write $h = \sigma(k)$, and so $d = \sigma(k)\sigma(a)\sigma(k)^{-1} = \sigma(kak^{-1}) \in \sigma(\mathcal{C})$. Thus, $\sigma(\mathcal{C})$ is indeed the whole conjugacy class. ■

In particular, automorphisms that are conjugations must fix all conjugacy classes. In the case of the symmetric group, this is an if and only if.

Proposition. Let σ be an automorphism of S_n . If σ sends 2-cycles to 2-cycles, then σ is conjugation by some element of S_n .

Proof. We will find $a, b_2, \dots, b_n \in [n]$ such that $\sigma(1i) = (ab_i)$. Then σ is the same as conjugation by

$$1 \mapsto a \quad 2 \mapsto b_2 \quad \dots \quad n \mapsto b_n$$

on the elements $(1i)$, and so must be the same as conjugation by this map on all of S_n , since these elements generate S_n .

First, we'll show that the transpositions $\sigma(1i)$ and $\sigma(1j)$ are not disjoint. If they were, then $\sigma(1i)\sigma(1j)$ would have order 2. However,

$$\sigma(1i)\sigma(1j) = \sigma((1i)(1j)) = \sigma(ji1)$$

has order 3.

Now, the way we could have a problem is if $\sigma(1i) = (ab)$, $\sigma(1j) = (bc)$, and $\sigma(1k) = (ac)$, since this means that there isn't a shared element among all the $\sigma(1i)$. But in this case

$$\sigma(1i)\sigma(1j)\sigma(1k) = (ab)(bc)(ac) = (bc)$$

has order 2, while

$$\sigma((1i)(1j)(1k)) = \sigma(1kji)$$

has order 4. Thus, this is also impossible, and so we can find such $a, b_2, \dots, b_n$ and σ is inner. ■

So, putting everything together, we'll show the following.

Theorem. *Let $n \geq 3$ and $n \neq 6$. Then $\text{Aut}(S_n) \cong S_n$.*

Proof. Let σ be an automorphism of S_n that is not inner. Then it doesn't send the conjugacy class of 2-cycles to itself. Since the image of a 2-cycle must still have order 2, this means that σ must send the conjugacy class of 2-cycles to the conjugacy class of k disjoint 2-cycles. So it suffices to show that this is impossible.

We'll count the number of 2-cycles and the number of permutations with k disjoint 2-cycles and show that these cannot be equal. There are $\binom{n}{2}$ 2-cycles, and $\frac{n!}{2^k \cdot k! \cdot (n-2k)!}$ products of k disjoint 2-cycles. So these are equal if and only if

$$k! \cdot 2^{k-1} = (n-2)(n-3) \cdots (n-2k+1).$$

There are $2(k-1)$ terms in both products (not counting the 1 in $k!$). If $n-2k+1 \geq 2$, this then means that the RHS is strictly greater than the LHS (since the LHS always has at least 2 factors of 2 in the product - one from 2^{k-1} and one from $k!$). Thus, we need to consider the case $n-2k+1 = 1$, so that $n = 2k$. We're trying to solve $k! \cdot 2^{k-1} = (2k-2)!$, or

$$2^{k-1} = (2k-2)(2k-3) \cdots (k+1).$$

We can check that this doesn't work for $k = 2$ and that we do have equality for $k = 3$. For $k \geq 4$, we have at least two terms in the RHS, and so the RHS is always divisible by an odd number $\neq 1$, while the LHS is always a power of 2. Thus, the only solution is $n = 6$ and $k = 3$, and so for $n \neq 6$ we see that $\text{Aut}(S_n) \cong S_n$, as desired. ■

It turns out that $[\text{Aut}(S_6) : S_6] = 2$.